

Anish Acharya

Website [Google Scholar](#) [Linkedin](#) [GitHub](#)

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Education

- **University of Texas, Austin** Austin, TX
PhD, Electrical and Computer Engg. Fall, 2019 – Spring, 2025
Advisor(s): Prof. Inderjit S Dhillon and Prof. Sujay Sanghavi
PhD Committee: Prof. Alex Dimakis, Prof. Ludwig Schmidt, and Prof. Nhat Ho
PhD Thesis: **Robust and Efficient Learning in High Dimensions from Noisy Data**
- **University of California, Irvine** Irvine, CA
M.S. Electrical Engg. and Computer Science Fall, 2013 – Winter, 2014
- **Jadavpur University** Kolkata, India
B.E. Electronics and Instrumentation Engineering Spring, 2009 – Spring, 2013
Advisor(s): **Amitava Gupta and Shantanu Das**

Core Technical Skills

- **Research Interests:** My recent research focuses on developing theoretically grounded, robust (e.g to distribution shifts, label noise, byzantine corruption) and (parameter / data) efficient training and fine-tuning algorithms for Foundation Models (e.g., diffusion models, LLMs). We study these problems under (un/ weakly / semi / fully) supervised settings.
Beyond foundational training methods, I work on multi-agent LLM architectures and develop core methods such as prompt optimization, controllable generation, self-play dialog simulation, formal state modeling, LLM-as-Judge evaluation – deploying these systems in production with rigorous evaluation loops.
- **Keywords:** Large Language Models, Multi-Agent Framework, Decentralized AI, Deep Learning, Machine learning, Robust (outliers, distribution shifts) Training, Data / Parameter Efficient Training, Self Supervised Learning, Diffusion Models, Federated Learning, Natural Language Processing.

Industry Experience

- **Amazon** Austin, TX
Senior Research Scientist (Full Time) April 2025 -
Building Decentralized AI Marketplace.
My work spans the full life-cycle of **multi-agent dialog system** – defining the architecture, developing core methods (prompt optimization, controllable generation, self-play for dialog simulation, formal state and artifact modeling, LLM-as-Judge evaluation), integrating these into production-ready pipelines, and closing the loop through evaluation and iteration.
This research directly powered the launch of **AWS Marketplace Agent Mode** at re:Invent 2025 – a **deep research** tool bringing structured requirements analysis, long-context reasoning, and grounded solution comparison to real customers.
In the news : **Amazon Blog, The Ravit Show, CRN, ZDNet, Inside Partnering**
- **Eidon AI** Austin, TX
Head of AI Research (Part Time) July 2024 - Feb 2025
The AI is the Blockchain, the Blockchain is the AI.
I proposed and developed a novel **multi-agent dynamic point-based system** that integrates blockchain technology with AI – making it accessible, transparent, and equitable for everyone. Designed and implemented comprehensive evaluation metrics for content quality, AI alignment, and community standards, establishing a flexible **tokenomics framework for fair reward distribution** in our decentralized network. Spearheaded the end-to-end system architecture, successfully launching the platform. Take a look at the **Technical Docs** for details on the algorithms.
- **Meta** Menlo Park, CA
Student Researcher (Part Time) Fall, 2022
Research Scientist (Intern) Summer, 2022
Student Researcher (Part Time) Fall, 2021
Research Scientist (Intern) Summer, 2021
With Meta Capacity Engineering and Meta Fundamental AI Research group, I worked on online low resource training to scale up Facebook's recommendation models. In particular, we investigated contrastive learning applied to weakly supervised prediction task.
- **Microsoft** Redmond, WA
Research Scientist (Intern) Summer, 2020
As part of the **Speech and Dialog Research Group**, my research was focused on developing algorithms to train large scale robust models in Federated Learning setting. Also, designed an **Open Source Federated Learning Platform - OpenMSFTL**.

- Amazon** Sunnyvale, CA
Applied Scientist (Full Time) June 2016 - July 2019
 As an Applied Scientist in the **Amazon Alexa AI** group, I was involved in proposing, designing and implementing NLP models shared across all alexa devices. I was also a member of the **Alexa Conversational AI** where we launched **Alexa Conversations**. I was involved in designing and implementing the Natural Language Processing (NLP) frameworks for the following Alexa Domain Models: Timers, Alarms, Reminders, Calendar, Communications consumed across multiple Alexa IOT devices (e.g. Echo Show, FireTV). I was also the **primary inventor** of the following research: **Model Compression via Low Rank Matrix Factorization (paper, patent)**, **Goal Oriented Dialog Generation (paper, patent)**. My work at Amazon has been featured at: **Forbes**, **CNN**, **Packt Blog**, **VentureBeat**, **HakerNews**, **Amazon Science (2019)**, **Amazon Blog (2019)**, **Amazon Science (2021)**.
- eBay inc** San Jose, CA
Data Scientist (Contract) Nov 2015 - June 2016
 As part of the Customer Insights and Analytics (CSI) group, I worked on developing **deep recommendation algorithms** to **personalize buyer search** and *personalized eBay advertising* e.g. **generative models for targeted (personalized) emails**.
- Schlumberger** Houston, Tx
Data Scientist (Full Time) March, 2015- Nov 2015
 As one of the early member in the newly formed **Digital Transformations Services** AI research group some of my work include: **Failure Prediction** of deployed oil service instruments, **Oil Level Estimation** from various sensor data generated from different parts of oil extraction life-cycle.
- FEM inc (acquired by Nielsen Gracenote)** Los Angeles, CA
Data Engineer (Full Time) Winter, 2015
 I worked closely with the co-founders on envisioning Natural Language Processing models, designing content recommendation system using these models and building visualization tools for interactive online analytics dashboard.
- Toyota Technological Institute Chicago (TTIC)** Chicago, IL; Ann Arbor, MI
Visiting Researcher (Intern) Summer, 2014
 We developed a real-time stereo algorithm to reconstruct 3D surrounding for autonomous car. Another extension was to develop a stereo evaluation benchmark based on slanted plane model of the surrounding. This project was developed for **Toyota Research in North America (TRINA)**.

Teaching Experience

- University of Texas at Austin** Austin, TX
 - Directed Reading Program (DiRP): **Natural Language Optimization (NLO) for LLMs** Fall, 2025
 - MIS 285N: **Introduction to Deep Learning** with Prof. C. Caramanis, Prof. A. Dimakis Spring, 2024
 - MIS 185N: **Emerging Trends in Machine Learning** with Prof. S. Shakkottai Spring, 2024
 - ECE 461P: **Data Science Principles** with Prof. Sujay Sanghavi. Fall, 2023
- University of California Irvine** Irvine, CA
 - EECS 50 : Discrete Time signals and systems (Undergraduate) Fall, 2014
 - EECS 280A: Advanced Engineering Electromagnetics I (Graduate) Spring, 2014
 - EECS 118: Introduction to Knowledge Management and software Engineering (Undergraduate) Winter, 2014
 - EECS 119: Introduction to VLSI (Undergraduate) Fall, 2013

Academic Awards

- Outstanding Reviewer** award at ICML 2026. Spring, 2026.
- Meta PhD Research Grant** of \$ 100,000 Spring, 2021 - Spring, 2023.
- Meta PhD Research Fellowship Finalist** - AI HW/SW co-design (**Press Release**) Spring, 2022.
- Travel Grants** to attend ML Conferences (NeuRips, ICML, AISTATS, UAI) from UT Austin. Spring, 2022-Fall, 2024.
- Travel Grants** to visit Simons Institute for the theory of computing, UC Berkley. Fall 2021, 2022.
- Awarded Bachelor of Engineering with **Highest Honors (First Class)**, Jadavpur University Fall, 2013
- INSPIRE Fellowship, Govt. of India.** Fall, 2009 - Spring, 2013.
- Fellowship , Ministry of Human Resource Development, Govt. of India** for Academic Excellence. Fall, 2009 - Spring, 2013.
- Mamraj Agarwal Rashtriya Puraskar (National Award)** from Governor (West Bengal), for Academic Excellence. Summer, 2007.

Public Engagements

- Spring 2026. (Remote) AWS Marketplace and Partner Services - Prompt Optimization (Invited Talk)
- Winter 2026. (Austin, TX) AWS AI Training - Evaluations (Instructor)
- Fall 2025. (Seattle, WA) Amazon Fulfillment Simulation Symposium (AFSS), 2025 (Invited Talk)
- Summer 2025. (Vancouver, CAN) **ICML 2025** (Recorded Talk, Poster)
- Spring 2024. (Austin, TX) Special Topics in Machine Learning: Generative Models, UT Austin. (Invited Talk)

- Fall 2023. (Austin, TX) **Future of Chatbots and Conversational AI Summit (FOCA 2023)** (Keynote, Panel)
- Fall 2022. (Virtual) Meta: Fundamental AI Research (FAIR) (Invited Talk)
- Fall 2022. (Berkley, CA) **Simons Institute for the Theory of Computing: Data-Driven Decision Processes Workshop** (Poster)
- Fall 2022. (Virtual) Meta: Ads ML Research (Invited Talk)
- Summer 2022. (Eindhoven, Netherlands) **UAI 2022** (Spotlight, Poster)
- Summer 2022. (Baltimore, MD) **ICML 2022** (Recorded Talk, Poster)
- Spring 2022. (Austin, TX) **6g@UT Forum** (Poster)
- Spring 2022. (Virtual) **AISTATS 2022** (Recorded Talk, Poster)
- Spring 2022. (Virtual) **University of Exeter - Department of Mathematics** (Invited Talk)
- Fall 2021. (Virtual) University of Denver - Department of Computer Science (Guest Lecture)
- Fall 2021. (Berkley, CA) **Simons Institute for the Theory of Computing - IFML/CCSI Symposium** (Poster)
- Fall 2021. (Virtual) **Meta: Fundamental AI Research (FAIR)** (Invited Talk)
- Summer 2021. (Virtual) **Meta : Capacity Engineering Research** (Invited Talk)
- Spring 2021. (Virtual) **NSF-TRIPODS Workshop on Communication Efficient Distributed Optimization** (Poster)
- Summer 2020. (Virtual) **Microsoft: Azure AI** (Invited Talk)
- Fall 2019. (Austin, Tx) **UT Austin Graduate Research Showcase** (Poster)
- Fall 2019. (Austin, Tx) **Texas Wireless Summit** (Poster)
- Spring 2019. (Honolulu, Hawaii) **AAAI 2019** (Oral)
- Summer 2018. (Seattle, WA) **Amazon Machine Learning Conference** (Oral)

Granted US Patents (Primary Inventor)

1. (Amazon Alexa) *"Goal Oriented Dialog Generation using Dialog Template, API and Entity data"*; July, 2022. [USPTO Link](#)
A. Acharya , A. Metallinou, T. Chung, S. Paul, S. Chandra, C. Lin, D. H. Tur, A. Mandal
2. (Amazon Alexa) *"Natural Language Processing : Amazon Alexa AI"*; Dec 2020. [USPTO Link](#)
A. Acharya , A. Metallinou, R. Goel, I. Dhillon

Publications

Preprints

1. *"SPLICE: Structured Prompt Local Iterative Combinatorial Evolution"*
Anish Acharya, Phillip Studans, Amit Dhanda, Ninad Vivek Rao, Vishal Amarlal Khatri, Sina Amini Niaki, Brian Verkhovsky
2. *"RoPoLL: Robust Panel of LLM Judges"*
Anish Acharya, Kris Pan, Brian Verkhovsky
Short version accepted at [International Conference on Machine Learning \(ICML\), 2026](#) : Statistical Frameworks for Uncertainty in Agentic Systems (AgenticUQ) workshop. [OpenReview](#)
3. *"Understanding Contrastive Representation Learning from Positive Unlabeled (PU) Data."* [Preprint](#)
Anish Acharya, Li Jing, Bhargav Bhushanam, Dhruv Choudhary, Michael Rabbat, Sujay Sanghavi, Inderjit S Dhillon
Short version presented at [International Conference on Machine Learning \(ICML\), 2022](#) : PODS Workshop. [OpenReview](#)
Short version presented at [Simons Institute - UC Berkley](#) : Data-Driven Decision Processes Workshop.

Peer Reviewed Journal Articles.

1. *"Neural Distributed Source Coding"*
[IEEE Journal on Selected Areas in Information Theory](#). 2024. [Read Online](#)
A. Nagle, J. Whang, **A. Acharya**, H. Kim, A. Dimakis.
2. *"On the Benefits of Multiple Gossip Steps in Communication Constrained Federated Learning"*
[IEEE Transactions on Parallel and Distributed Systems](#), 2021. [Read Online](#)
A. Hashemi, **A. Acharya**, R. Das, H. Vikalo, S. Sanghavi, I. Dhillon.
3. *"Extending The Concept of Analog Butterworth Filter For Fractional Domain"*
[Signal Processing \(Elsevier \)](#), 2014. [Read Online](#)
Anish Acharya, Saptarshi Das, Indranil Pan, Shantanu Das.
4. *"Simulation studies on the design of optimum PID controllers to suppress chaotic oscillations in a family of Lorenz-like multi-wing attractors"*
[Mathematics and Computers in Simulation \(Elsevier \)](#), 2014. [Read Online](#)
Saptarshi Das, **Anish Acharya**, Indranil Pan

Peer Reviewed Conference Articles.

1. "Geometric Median Matching for Robust k -Subset Selection from Noisy Data".
[International Conference on Machine Learning \(ICML\), 2025](#) [Read Online](#)
Anish Acharya, Sujay Sanghavi, Alex Dimakis, Inderjit Dhillon.
Short version presented at [International Conference on Machine Learning \(ICML\), 2024 - FMWild Workshop](#).
 2. "Faster Non-Convex Federated Learning via Global and Local Momentum". (Spotlight)
[The Conference on Uncertainty in Artificial Intelligence \(UAI\) 2022](#) [Read Online](#)
R. Das, **A. Acharya**, A. Hashemi, S. Sanghavi, I. Dhillon, U. Topku.
 3. "Robust Training in High Dimensions via Block Coordinate Geometric Median Descent". (Spotlight)
[International Conference on Artificial Intelligence and Statistics \(AISTATS\) 2022](#). [Read Online](#)
A. Acharya, A. Hashemi, P. Jain, S. Sanghavi, I. Dhillon, U. Topku.
Short version presented at Joint IFML/CCSI Symposium (Simons Institute - UC Berkley).
Short version presented at NSF-TRIPODS Workshop on Communication Efficient Distributed Optimization.
 4. "LDKP: A Dataset for Identifying Keyphrases from Long Scientific Documents"
[ACM International Conference on Information and Knowledge Management \(CIKM\) 2022](#)
DL4SR'22: Workshop on Deep Learning for Search and Recommendation. [Read Online](#)
D Mahata, N Agarwal, D Gautam, A Kumar, S Parekh, Y K Singla, **A Acharya**, R R Shah.
 5. "Alexa Conversations: An Extensible Data-driven Approach for Building Task-oriented Dialogue Systems".
[Conference of the North American Chapter of the Association for Computational Linguistics \(NAACL\) 2021](#). [Read Online](#)
A. Acharya, S. Adhikari, S. Agarwal, V. Auvray, N. Belgamwar, A. Biswas, S. Chandra, T. Chung, M. Zarandi, R. Gabriel, S. Gao, R. Goel, D. Hakkani-Tur, J. Jezabek, A. Jha, J. Kao, P. Krishnan, P. Ku, A. Goyal, C. Lin, Q. Liu, A. Mandal, A. Metallinou, V. Naik, Y. Pan, S. Paul, V. Perera, A. Sethi, M. Shen, N. Strom and E. Wang;
[Media: CNN, Forbes, VentureBeat, Amazon Science, Alexa Skill](#).
 6. "GupShup: An Annotated Corpus for Abstractive Summarization of Open-Domain Code-Switched Conversations".
[Conference on Empirical Methods in Natural Language Processing \(EMNLP\) 2021](#). [Read Online](#)
L. Mehnaz, D. Mahata, A. Kumar, U. S. Gunturi, R. Jain, G. Gupta, R. Gosangi, I. G. Lee, **A. Acharya**, R. Shah.
 7. "Online Embedding Compression for Text Classification using Low Rank Matrix Factorization". (Spotlight)
[AAAI Conference on Artificial Intelligence \(AAAI\) 2019](#). [Read Online](#)
Anish Acharya, Rahul Goel, Angeliki Metallinou, Inderjit Dhillon.
[Media: Amazon Science, Packt](#)
 8. "Stability Analysis Of Delayed System Using Bode's Integral".
[International Conference on Computer Communication and the Internet \(ICCCI\) 2013](#) [IEEE Xplore](#)
A. Acharya, D. Mitra, K. Halder
 9. "Optimum PID Control of Multi-wing Attractors in A Family of Lorenz-like Chaotic Systems"
[International Conference on Computing, Communication and Networking Technologies \(ICCCNT\) 2012](#). [IEEE Xplore](#)
A. Acharya, S. Das, I. Pan
 10. "Optimized Quality Factor of Fractional Order Analog Filters with Band-Pass and Band-Stop Characteristics"
[International Conference on Computing, Communication and Networking Technologies \(ICCCNT\) 2012](#). [IEEE Xplore](#).
A. Pakhira, S. Das, **A. Acharya**, I. Pan, S. Saha
 11. "Identification of nonlinear systems from the knowledge around different operating conditions: a feed-forward multilayer ANN based approach"
[International Conference on Parallel, Distributed and Grid Computing \(PDGC\) 2012](#) [IEEE Xplore](#)
S. Saha, S. Das, **A. Acharya**, A. Kumar, S. Mukherjee, I. Pan, A. Gupta
 12. "Least square and instrumental variable system identification of AC servo position control system with fractional Gaussian noise"
[International Conference on Energy, Automation and Signal \(ICEAS\) 2011](#) [IEEE Xplore](#)
S. Das, A. Kumar, I. Pan, **A. Acharya**, S. Das, A. Gupta.
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Academic Services

- Reviewer - ([JMLR](#)) [Journal of Machine Learning Research](#)
- Reviewer - ([ACL](#)) [Association for Computational Linguistics](#)
- Reviewer - ([ISIT](#)) [IEEE International Symposium on Information Theory](#)
- Reviewer - ([ICLR](#)) [International Conference on Learning Representations](#)
- Reviewer - ([UAI](#)) [Conference on Uncertainty in Artificial Intelligence](#)
- Reviewer - ([ICML](#)) [International Conference on Machine Learning](#)
- Reviewer - ([AISTATS](#)) [Conference on Artificial Intelligence and Statistics](#)
- Reviewer - ([NeurIPS](#)) [Conference on Neural Information Processing Systems](#)
- Reviewer - [IEEE Transactions on Knowledge and Data Engineering](#)

- Reviewer - **Pattern Recognition Letters (Elsevier)**
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References

- **Inderjit S Dhillon**
Gottesman Family Centennial Professor of Computer Science at UT Austin
Vice President and Google Fellow, Google.
ACM Fellow, IEEE Fellow, SIAM Fellow, AAAS Fellow, SIAM Linear Algebra Prize.
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